

Hypothesis testing with e-values

Chapter 5: Universal inference

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Recap

Relevant concepts:

- **Validity (Chapter 1)**: E e-variable for $\mathcal{P}$ iff $\mathbb{E}_{\mathbb{P}}[E] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$
- **Markov's inequality (Chapter 2)**: $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$ for all $\mathbb{P} \in \mathcal{P}$
- **Randomised Markov's inequality (Chapter 2)**: $\mathbb{P}(E \geq U/\alpha) \leq \alpha$ for all $\mathbb{P} \in \mathcal{P}$ where $U \sim \text{Unif}[0, 1] \perp E$
- **Efficiency (Chapter 3)**: e-power against $\mathbb{Q} \in \mathcal{Q}$ is $\mathbb{E}_{\mathbb{Q}}[\log E]$
- E-variable **construction** (assume all $\mathbb{P}, \mathbb{Q} \ll \mathbb{L}$ to define p, q):
 - **Likelihood ratio (Chapter 1)**: $E = q(X)/p(X)$ log-optimal for $\mathbb{P}$ vs $\mathbb{Q}$
 - with a few extensions to $\mathcal{P}$ vs $\mathcal{Q}$ under specific assumptions (e.g. monotonicity) (Chapter 2)
 - Method of **mixtures (Chapter 3)**: $E = \int \prod_{i=1}^n (q(X_i)/p(X_i)) \nu(dq)$ for $\mathbb{P}$ vs $\mathcal{Q}$
 - **Plug-in method (Chapter 3)**: $E = \prod_{i=1}^n q_{i-1}(X_i)/p(X_i)$, with $q_{i-1} \in \mathcal{Q}$ predictable, for $\mathbb{P}$ vs $\mathcal{Q}$

Motivation

Irregular problems are settings where general methods to obtain (asymptotic) level- α tests break:

- MLE not asymptotically normal $\implies$ lose standard **z-test**
- Wilks' theorem not valid $\implies$ lose **LRT**
- Fisher information matrix not invertible $\implies$ lose both
- Bootstrap without guarantees $\implies$ lose **resampling-based tests**

Universal inference provides (**non-asymptotic**) level- α tests for $\mathcal{P}$ vs $\mathcal{Q}$ under **no regularity assumptions**, except for (**very** mild):

- **Densities** p, q wrt common $\mathbb{L}$ for all $\mathbb{P} \in \mathcal{P}, \mathbb{Q} \in \mathcal{Q}$
- **Data** $X^n = (X_1, \dots, X_n)$ **i.i.d** from $\mathbb{P}^*$

Universal inference (UI)

Given $\{\mathcal{P}, \mathcal{Q}, X^n\}$, define the **split likelihood ratio** (split-LR) e-variable:

1. **Split** X^n into D_0, D_1 based on $Z_1, \dots, Z_n \stackrel{iid}{\sim} \text{Bern}(\pi_0) \perp X^n$
2. **Pick alternative** $\hat{q}_1 \in \mathcal{Q}$ based on D_1 (e.g. MLE, **Bayes**, etc.)
3. **Compute MLE** $\hat{p}_0 \in \mathcal{P}$ based on D_0
4. Define

$$E = \frac{\prod_{i \in D_0} \hat{q}_1(X_i)}{\prod_{i \in D_0} \hat{p}_0(X_i)}$$

Properties:

- **Validity**: E is an e-variable by $D_0 \perp D_1$, tower property, MLE optimality, and validity of LR
- **Split-LRT**: valid α -level test by Markov's inequality
- Example of **plug-in** e-variable: $q_{i-1} = \hat{q}_1$ vs $p = \hat{p}_0$
- Can be combined with **method of mixtures**

Universal confidence sets

Duality between α -level tests and $1 - \alpha$ confidence sets:

- Parametric setting: $\{p_\theta : \theta \in \Theta\}$ with $X^n \stackrel{iid}{\sim} p_{\theta^*}$
- As before, estimate $\hat{\theta}$ based on D_1
- Define

$$C_\alpha(X^n) = \left\{ \theta \in \Theta : \prod_{i \in D_0} \frac{p_{\hat{\theta}}(X_i)}{p_\theta(X_i)} \leq \frac{1}{\alpha} \right\} = \left\{ \theta \in \Theta : \frac{\mathcal{L}_0(\hat{\theta})}{\mathcal{L}_0(\theta)} \leq \frac{1}{\alpha} \right\}$$

- Non-asymptotic $1 - \alpha$ coverage by Markov's inequality

$$\mathbb{P}_{\theta^*}(\theta^* \in C_\alpha(X^n)) \geq 1 - \alpha$$

- This construction is central to e-values and not specific to UI

Quick aside: standard LRT confidence set

Standard LRT-based $1 - \alpha$ confidence set:

- Compute MLE $\hat{\theta}$ on X^n
- Define

$$\begin{aligned} C_{\alpha}^{\text{LRT}}(X^n) &= \left\{ \theta \in \Theta : 2 \log \left(\frac{\mathcal{L}_n(\hat{\theta})}{\mathcal{L}_n(\theta)} \right) \leq \chi_{d,1-\alpha}^2 \right\} \\ &= \left\{ \theta \in \Theta : \frac{\mathcal{L}_n(\hat{\theta})}{\mathcal{L}_n(\theta)} \leq \exp \left(\frac{\chi_{d,1-\alpha}^2}{2} \right) \right\} \end{aligned}$$

- Asymptotic $1 - \alpha$ coverage by Wilks' theorem
- Trade-off need for data splitting vs regularity conditions (and asymptotic guarantees)

Simulations - Optimal split

For $X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta^*, I_d)$, one can compute the π_0^* that **minimises the expected squared radius** of the split-LRT confidence set:

- As $d \rightarrow \infty$, $\pi_0^* \rightarrow 1/2$
- As $\alpha \rightarrow 0$, $\pi_0^* \rightarrow 1$

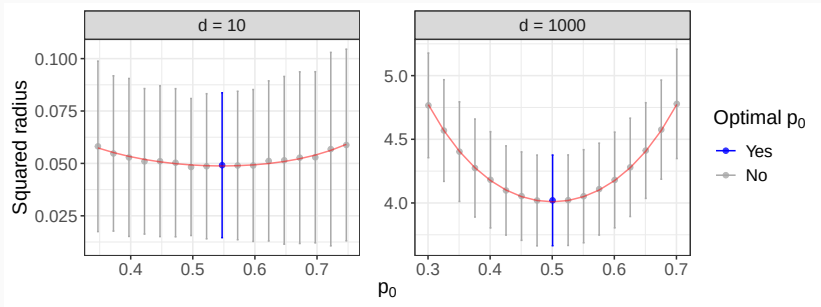


Figure 1: Effect of changing π_0 with $n = 1000$, $\theta^* = 0$, $\alpha = 0.10$ and varying π_0 .

Simulations - Comparison with LRT

In the same setting, one can compute the **limit of the expected squared radius** of split-LRT vs LRT confidence sets.

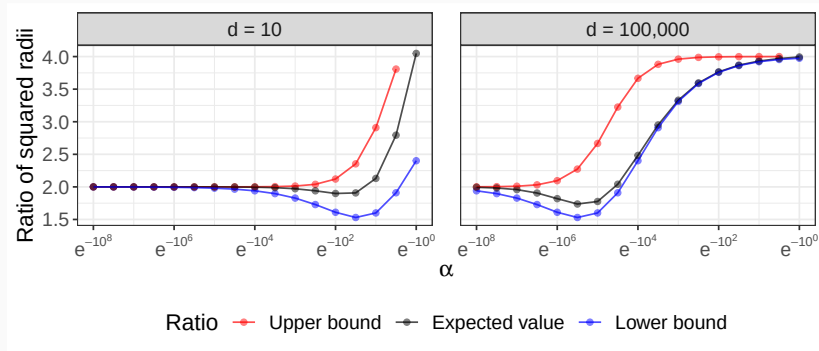


Figure 2: Bounds for $\mathbb{E} [r^2(C(X^n))/r^2(C^{\text{LRT}}(X^n))]$.

Subsampled likelihood ratio

- Split-LR e-variables introduces **variability** due to random split
- Solve by **averaging** over multiple splits
 - **Crossfit-LR** e-variable:

$$E' = (E_0 + E_1)/2$$

- **Subsampled-LR** e-variable: use B **identical independent** splits

$$\bar{E}^{(B)} = \frac{1}{B} \sum_{b=1}^B E^{(b)}$$

- **Validity** of e-variables follows from **linearity of expectation**
- **Efficiency** (weakly) improves with subsampled-LR by Jensen's

$$\mathbb{E}_{\mathbb{Q}}[\log \bar{E}^{(B)}] \geq \mathbb{E}_{\mathbb{Q}}[\log E^{(1)}]$$

- **Markov's** inequality can be used to obtain valid tests and confidence sets when B is **fixed in advance**, however...

Exchangeable Markov's inequality

- By construction, $E^{(1)}, E^{(2)}, \dots$, are **exchangeable**
- This implies that $\bar{E}^{(B)}$ is a **backward martingale**
- **Exchangeable Markov's inequality**:

$$\mathbb{P} \left(\exists B \geq 1: \bar{E}^{(B)} \geq \frac{1}{\alpha} \right) \leq \alpha$$

- **Randomised exchangeable Markov's inequality**:

$$\mathbb{P} \left(E^{(1)} \geq U/\alpha \text{ or } \exists B \geq 1: \bar{E}^{(B)} \geq \frac{1}{\alpha} \right) \leq \alpha$$

- Key for proof is **backward**-variant of **Ville's inequality** (**Chapter 7 spoiler**)
- This allows to **monitor** $\bar{E}^{(B)}$ as B increases and **stop** as soon as $\bar{E}^{(B)} \geq 1/\alpha$

Simulations - Variability

In the previous setup, LRT gives an exact $1 - \alpha$ confidence set.

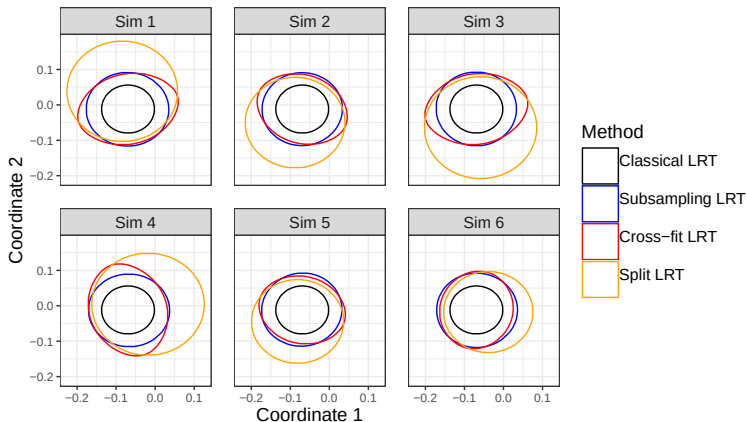


Figure 3: Coverage regions for $n = 1000$, $d = 2$, $B = 100$, $\alpha = 0.1$.

Simulations - Power for multivariate Gaussian

In the usual setup, **power** difference increases as d increases.

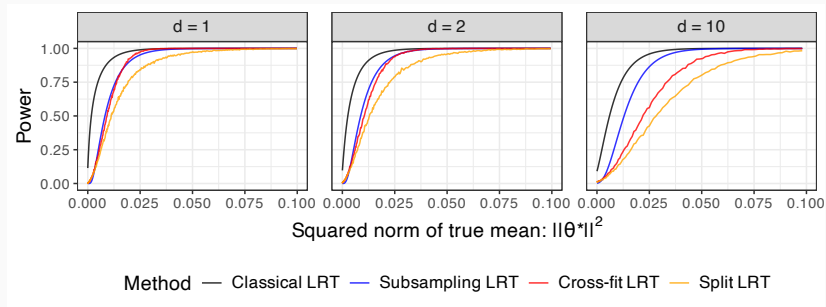


Figure 4: Empirical power when testing $H_0 : \theta^* = 0$ versus $H_1 : \theta^* \neq 0$ across varying true $\|\theta^*\|^2$.

Simulations - Power for Gaussian mixture

When $X_1, \dots, X_n \stackrel{iid}{\sim} 0.25N(\mu_1, 1) + 0.75N(\mu_2, 1)$, the LRT is not available.

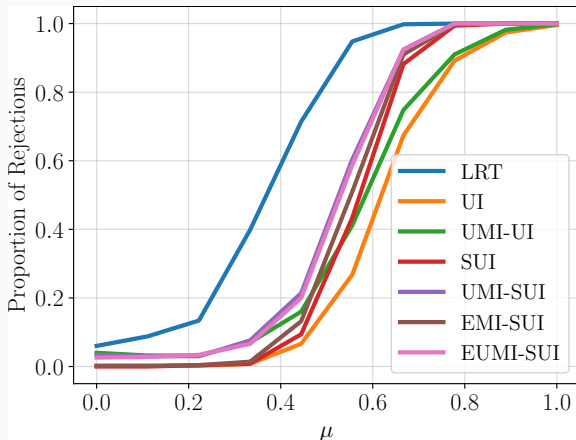


Figure 5: Empirical power when testing $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$ for $n = 500$ for varying $-\mu_1 = \mu_2 \in [0, 1]$

UI relies on defining/maximising a suitable **likelihood**:

- **Nuisance** parameters via the **profile** likelihood:

$$\mathcal{L}_0^\dagger(\psi) = \sup_{\theta: g(\theta)=\psi} \mathcal{L}_0(\theta)$$

- **Unbounded** likelihood via **smoothing**:

$$\begin{aligned}\tilde{p}_\theta(y) &= \int k(x, y) p_\theta(x) dx \\ \tilde{\mathcal{L}}_0(\theta) &= \prod_{i \in D_0} \exp \left(\int k(X_i, y) \log \tilde{p}_\theta(y) dy \right)\end{aligned}$$

- **Non-i.i.d.** data via **conditional** likelihood given D_1

UI relies on defining/maximising a suitable **likelihood**:

- **Expensive** likelihood via **relaxation**:

$$F_0: \max_{\theta} F_0(\theta) \geq \max_{\theta} \mathcal{L}_0(\theta)$$

$$\hat{\theta}_0^F = \arg \max_{\theta} F_0(\theta)$$

- **Misspecified** likelihood via **power** likelihood:

$$\mathcal{L}'_0(\theta) = \mathcal{L}_0(\theta)^\eta \text{ for } \eta \in (0, 1)$$

Takeaways

Summary:

- Universal inference reduces valid $\mathcal{P}$ vs $\mathcal{Q}$ testing to **MLE estimation** in $\mathcal{P}$ via **likelihood ratios**
- **Splitting** helps efficiency by picking data-driven alternative in $\mathcal{Q}$
- **Subsampling** improves power and variability
- **Confidence sets** can be constructed by test inversion
- **Uniform-time** guarantees by averaging exchangeable e-variables